

# Maximum Nullity of Cyclic Covers of Graphs

A monodromy ceiling, and the zero forcing  
number of  $P(n, k)$

**Research Binder**

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Compiled 28 September 2026

## Overview & Standards of Proof

This research binder contains the complete, dated record of the investigation into exact zero forcing numbers and maximum nullity of generalized Petersen graphs  $P(n, k)$  and general cyclic covers.

Every claim in the paper is labelled by the standard it meets: **proved**, **proved by an exact certificate**, **proved by interval certification**, or **numerical**. Claims that have been refuted or withdrawn, including the author's own, are listed explicitly in the paper's Status section and dated in the log.

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# I. Official Abstract

(September 28, 2026)

Zero forcing is a colouring process: a coloured vertex with exactly one uncoloured neighbour forces that neighbour to colour. The zero forcing number  $Z(G)$  is the smallest starting set that colours everything, and it bounds the maximum nullity  $M(G)$  over symmetric matrices with the graph's pattern. For the generalized Petersen graphs  $P(n, k)$ , a 2020 paper claimed  $Z(P(n, 3))=8$  for  $n$  at least 12. A July 2026 note showed this is false and conjectured  $Z(P(n, 3))=8$  for  $n$  at least 13, identifying the missing ingredient as a lower-bound proof for all large  $n$ .

Eliminating the inner coordinates of any matrix carrying the  $P(n, k)$  pattern leaves one linear recurrence of order exactly  $2k+2$ . Its nullity is therefore at most  $2k+2$  the best known upper bound on  $Z$ . So the matrix method either settles  $Z$  completely or cannot settle it at all.

Reaching that ceiling with a rotation-invariant matrix provably confines  $n$  to one divisibility class containing at most one prime. Assembling it from tiles that each contribute the identity to the monodromy removes the restriction: tiles whose lengths sum to  $n$  settle  $n$ . I certified 77 such tiles, evaluating every Krawczyk bound in exact rational arithmetic rather than floating point.

That is the missing ingredient. Consequently  $Z(P(n, 2))$ ,  $Z(P(n, 3))$  and  $Z(P(n, 4))$  are determined for every  $n$ , the last two for the first time;  $Z(P(n, 3))=8$  for every  $n$  at least 13, proving Conjecture 5 of that note;  $Z(P(n, 4))=10$  for every  $n$  at least 18; and  $Z(P(n, 5))=12$  for every  $n$  at least 162.

## II. Research Plan

(September 28, 2026)

**Project.** Maximum Nullity of Cyclic Covers of Graphs: A Monodromy Ceiling and the Zero Forcing Number of Generalized Petersen Graphs **Category.** Mathematics **Finalist.** Aryan Padarathi

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### A. Rationale

Zero forcing began in two unrelated places: controllability of quantum spin systems, and placement of phasor measurement units in electrical grids. It was then connected to linear algebra by the inequality  $M(G) \geq Z(G)$ , where  $M(G)$  is the largest nullity attainable by a symmetric matrix whose off-diagonal support is the edge set of  $G$ . The inequality makes  $Z$  a combinatorial ceiling on a purely algebraic quantity, and makes lower bounds on  $Z$  obtainable by exhibiting a matrix.

The generalized Petersen graphs  $P(n, k)$  are a standard cubic test family. Rashidi, Shajareh Poursalavati and Tavakkoli (2020) proved  $Z(P(n, k)) \geq 2k+2$  and determined  $Z(P(n, 2))$ . Their Theorem 3.6, asserting  $Z(P(n, 3)) = 8$  for  $n \geq 12$ , is false: Krishnan's July 2026 correction note (arXiv:2607.19412) exhibits  $Z(P(12, 3)) = 7$ , the error being a case analysis that never excludes 7-vertex sets. That note leaves open the stabilization threshold and any rigorous lower bound for  $k \geq 4$ .

This project addresses those two open problems.

### B. Research question

1. Is there an intrinsic ceiling on what a matrix certificate can prove about  $Z(P(n, k))$ , and where does it sit relative to the known upper bound?
2. Can that ceiling be attained, and by an explicitly checkable matrix?
3. Can  $Z(P(n, k))$  be determined for **all**  $n$  rather than for sparse families of  $n$ ?

### C. Procedures

This is a theoretical mathematics project. There are no human participants, no vertebrate animals, no potentially hazardous biological agents, and no hazardous chemicals, activities or devices. All work is proof and computation performed by the finalist on a personal computer.

**C1 Structure.** Eliminate the inner coordinates from the kernel equations of an arbitrary matrix carrying the  $P(n, k)$  pattern and identify the resulting scalar recurrence, its order, and the leading coefficients. Establish when the nullity equals the recurrence order.

**C2 Construction.** For rotation-invariant matrices, reduce singularity to a polynomial condition on a discrete grid and construct matrices meeting the ceiling. Determine exactly when such a matrix is admissible.

**C3 Obstruction.** Determine whether the divisibility conditions arising in C2 are intrinsic, and prove the answer rather than observing it.

**C4 General  $n$ .** Construct matrices from repeated blocks whose contributions compose, so that one finite family of blocks settles an infinite set of  $n$ .

**C5 Certification.** Verify every constructed matrix at the highest standard available: exact integer or rational arithmetic where possible, and otherwise an interval (Krawczyk) contraction proof that a true real solution exists. Record which standard each claim meets.

**C6 Independent checking.** Compute  $Z$  directly by exhaustive search over vertex subsets for all sizes reachable, validate that solver against every published value, and reproduce each theorem numerically before relying on it.

### D. Data analysis

Results are theorems and explicit matrices, not measurements, so analysis is verification rather than statistics. Every computational claim is re-derived by a single script that reports pass or fail per check. Numerical searches are always

wrapped in a bound they cannot legally exceed (the ceiling of C1), so that an impossible answer is detected rather than believed. Where a search fails, the residual and the computational budget are both recorded, because a failure at a small budget is not evidence of impossibility.

## E. Risk and safety

None. No laboratory work, no fieldwork, no subjects of any kind.

## F. Bibliography

1. S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli. Computing the zero forcing number for generalized Petersen graphs. *J. Algebra Combin. Discrete Struct. Appl.* 7(2) (2020) 183193.
2. A. Krishnan. A correction to the zero forcing number of the generalized Petersen graphs  $P(n,3)$ . arXiv:2607.19412 (2026).
3. AIM Minimum Rank Special Graphs Work Group. Zero forcing sets and the minimum rank of graphs. *Linear Algebra Appl.* 428 (2008) 16281648.
4. J. H. Conway, A. J. Jones. Trigonometric diophantine equations. *Acta Arith.* 30 (1976) 229240.
5. I. Niven. *Irrational Numbers*. Carus Mathematical Monographs 11, MAA, 1956.
6. R. E. Moore, R. B. Kearfott, M. J. Cloud. *Introduction to Interval Analysis*. SIAM, 2009. (Krawczyk existence test.)

### III. Research Paper

(September 23–28, 2026)

The full paper follows this page. It states, for every claim, which of four standards it meets: proved; proved by an exact certificate; proved by interval certification; or numerical. Its Status section lists every claim of the author's own that has been refuted or withdrawn.